

ALTIRIG · PRECIS

AltiRig design precis

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DOCUMENT CONTROL

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Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/altirig		
Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.3	2026-10-03	Amish Chadha	TRL 3; numbers from ALR-CAL-001 and the constructable design (ALR-DDR-002)
0.2	2026-10-03	Amish Chadha	TRL 2 concept; vessel type, stand, density matching, safety design and media decided under Amish's 2026-10-03 pre-approval (ALR-DDR-001)
0.1	2026-09-30	Amish Chadha	Initial scaffold

AltiRig design precis

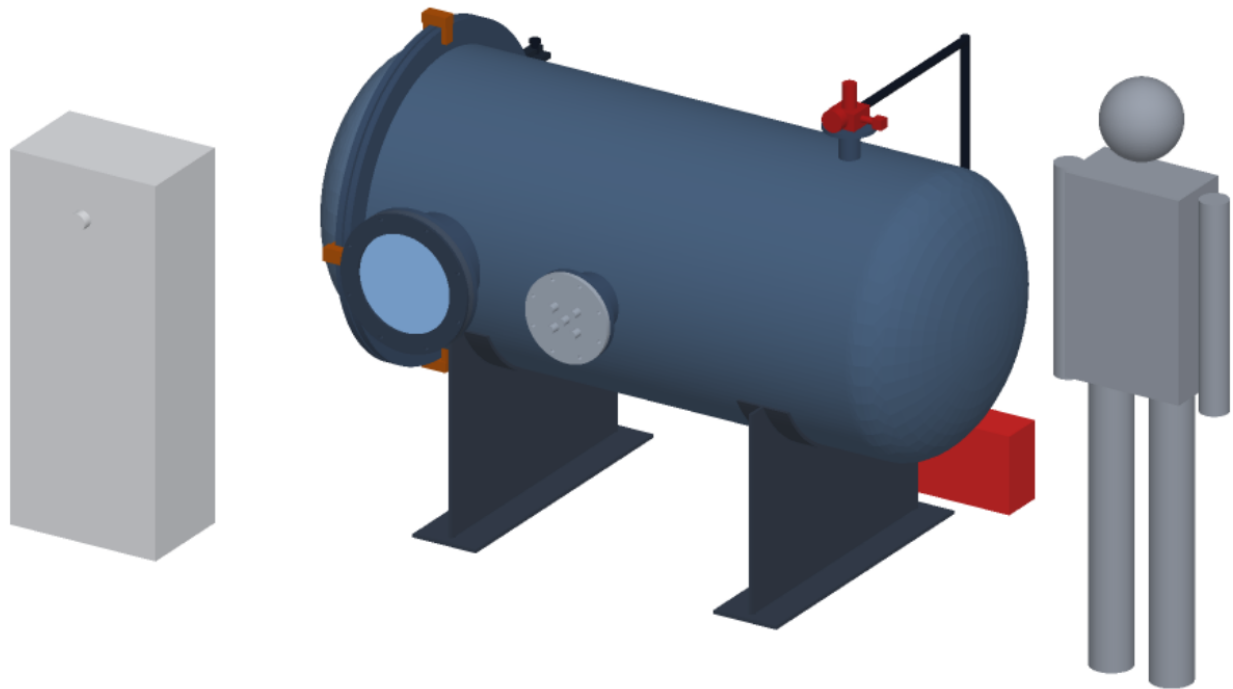
A test chamber that simulates thin air at 5,000 m so propellers and motors can be proven before going up the mountain.

AltiRig is a horizontal steel vacuum vessel, 813 mm in diameter and about 2 m long, with a flexure thrust stand inside. A vacuum pump lowers the pressure until the air inside has the density of 5,000 m (0.736 kg/m³, about 61.9 kPa at 20 °C), and the stand measures a propeller's thrust, torque and speed with the motor's voltage and current. The vessel is built from a steel pipe offcut and two bought tank heads, one of which is the door; air pressure holds the door shut while the chamber is under vacuum.

AltiRig

github.com/BoujeeEnjinia1701/altirig

CONCEPT, NOT FOR FABRICATION



Seen from the front right and above, 24 deg elevation. Grey figure: 1.75 m person for scale

Figure 1. AltiRig, seen from the front right and above: vessel on two saddles, door and latches at the left end, window and feed-through on the near side, valve manifold on top, pump behind, control cabinet in front.

■ How it works

The motor and propeller under test sit on the axis of the vessel, 510 mm in from the door, blowing toward the rear head. The operator closes the door, latches it and starts the pump. The controller reads two pressure sensors and two temperature probes inside, works out the air density, and holds it at the chosen value by switching the pump and opening a bleed valve that lets room air back in. As the motor runs, its heat warms the air; the controller raises the pressure a little to keep the density steady.

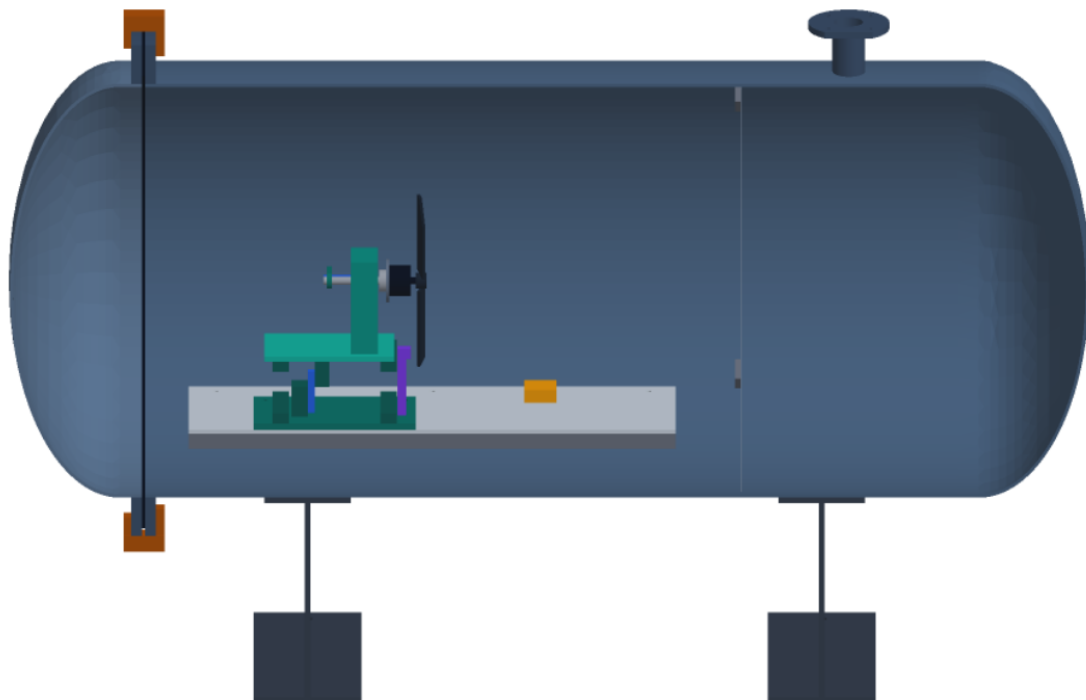
The propeller's thrust moves the carriage of the stand by about a tenth of a millimetre on two spring-steel leaves, and a load cell between the carriage and the base measures it. The motor is mounted on a shaft that turns freely in two bearings; its reaction torque presses an arm onto a second load cell. An optical sensor counts the blades. Motor power comes from a 1,500 W supply in the control cabinet through a speed controller and a sealed feed-through plate. Everything is logged to CSV at 20 Hz.

Inside a closed vessel the propeller drives a loop of air: down the middle, through a perforated baffle, round the rear head and back along the wall. To find what the loop does to the readings, each propeller is also run at ambient pressure and compared with an open-air stand; that comparison is published with the data.

AltIRig: cutaway

github.com/BoujeeEnjinia1701/altirig

CONCEPT, NOT FOR FABRICATION



Front half removed; seen from the front and above, 18 deg elevation

Figure 2. Cutaway with the window side removed: stand and motor on the deck at the door end, the propeller plane, the baffle and the rear head.

■ Components

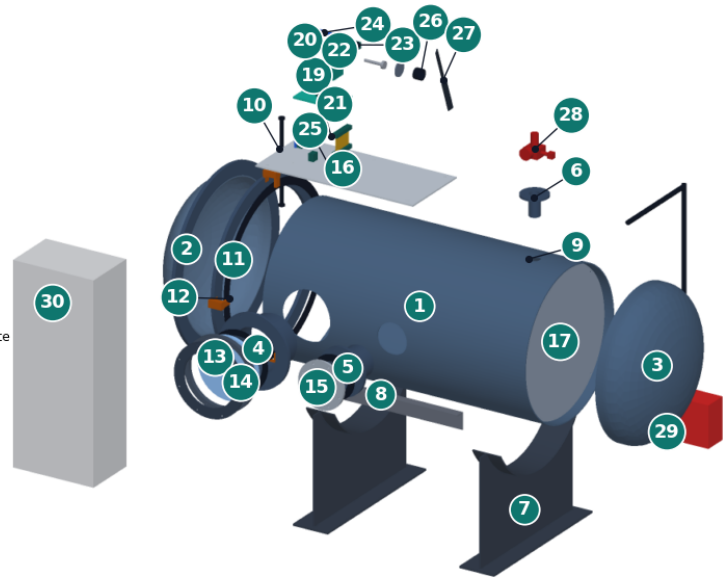
Table 1. Main components (numbers match `bom/bom.csv`).

#	Component	Role
1	Shell	813 x 6.35 mm steel pipe offcut, 1,500 mm long; the body of the vessel
2	Flange rings	20 mm steel rings on the shell end and on the door; the door seals on a sponge gasket between them
3	Dished heads	Two bought 2:1 ellipsoidal heads: one welded to the rear, one is the door
4	Window nozzle and flange	10 in heavy-wall stub with a flange for the window
5	Feed-through nozzle and flange	6 in heavy-wall stub with a flange for the cable plate
6	Manifold stub	2 in stub on top for the valve manifold
7	Saddles	Two welded steel saddles carrying the vessel at axis height 800 mm
8, 9	Deck rails and baffle tabs	Welded inside to carry the deck and the baffle
10	Hinge	Lugs and a 20 mm pin; the door swings to the far side
11, 12	Gaskets and latch clamps	EPDM gaskets; three latches pull the door on for the first seal
13, 14	Window, spacer and clamp rings	20 mm polycarbonate window, 340 mm, clamped on its flange
15	Feed-through plate	Aluminium plate with five potted IP68 cable glands
16, 17	Deck plate and baffle plate	Aluminium deck for the stand; 63 % open perforated baffle, removable
18 to 25	Thrust stand	Base, two flexure leaves and clamps, carriage, thrust cell, torque head on bearings, torque arm and cell, speed sensor
26, 27	Motor and propeller under test	Example 6S motor and 15 x 5 in propeller; users bring their own
28	Valve manifold	Relief valve (opens 55 kPa below atmosphere), gauge, normally-open bleed valve, vent and isolation valves
29	Vacuum pump and hose	170 L/min single-stage rotary vane pump
30 to 35	Cabinet, sensors, supply, speed controller, controller, safety controls	Mains protection, 1,500 W motor supply, logger, emergency stop, door interlock and arming key

AltiRig: exploded viewgithub.com/BoujeeEnjinia1701/altirig

CONCEPT, NOT FOR FABRICATION

- 1 Shell
- 2 Shell flange ring and Door flange ring
- 3 Rear head and Door head
- 4 Window nozzle and flange
- 5 Feed-through nozzle and flange
- 6 Manifold stub and flange
- 7 Saddle (door end) and Saddle (rear end)
- 8 Deck rail (far side) and Deck rail (window side)
- 9 Baffle tabs (4)
- 10 Hinge lugs on the shell ring and Hinge lugs on the door ring and Hinge pin
- 11 Door gasket and Window gasket and Feed-through gasket
- 12 Latch clamps (3)
- 13 Polycarbonate window
- 14 Window spacer ring and Window clamp ring
- 15 Feed-through plate with glands
- 16 Deck plate
- 17 Baffle plate (perforated)
- 18 Stand base plate (hidden in this view)
- 19 Flexure leaves (2) and Flexure clamps (4)
- 20 Thrust cell anchor block and Thrust cell hanger
- 21 Carriage plate
- 22 Torque head housing and Bearings (2) and Torque shaft and hub and Motor plate
- 23 Torque arm and Torque cell bracket
- 24 Thrust load cell and Torque load cell
- 25 Speed sensor post and sensor
- 26 Motor under test (example)
- 27 Propeller under test (example, 380 mm)
- 28 Valve manifold
- 29 Vacuum pump and Vacuum hose
- 30 Control and power cabinet
- 31 Air sensor box (hidden in this view)



Seen from the front right and above, 24 deg elevation; the stand, deck and test article lifted out above the vessel;
 numbers match bom/bom.csv

Figure 3. Exploded view; numbers match the bill of materials.

■ Key design choices

All were decided on 2026-10-03 under Amish's pre-approval and are argued in ALR-DDR-001 (TRL 2 review) and ALR-DDR-002 (design for construction).

- **Match density, not pressure.** Thrust and power depend on density. At room temperature the density of 5,000 m is reached at 61.9 kPa, not at the 54.0 kPa of the real atmosphere there. Pressure and temperature are logged so either can be reported.
- **A pipe vessel with tank heads, designed for full vacuum.** Thin drums collapse and flat panels need heavy ribs. A 6.35 mm pipe with dished heads has a collapse factor of 6.7 even at full vacuum, so a stuck relief valve cannot collapse it.
- **The door is a tank head held shut by the air.** At the 5,000 m point the air presses the door shut with 24 kN, so it cannot be opened until the chamber is vented.
- **A flexure stand, not rails.** Two spring-steel leaves carry the carriage with no sliding friction; the small share of thrust they carry is calibrated out.
- **Window and feed-through outside the propeller's fragment zone.** Both sit more than 18° from the propeller plane; the steel shell contains a broken blade.

- **No batteries inside the chamber in the first version.** The motor is powered from outside through the feed-through, so a battery fire cannot happen inside a closed vessel.

■ First-order numbers

From ALR-CAL-001; assumptions are stated there.

Table 2. First-order numbers.

Quantity	Value
Density range	Ambient down to 0.660 kg/m ³ (6,000 m) at up to 30 °C; relief at 46.3 kPa
Free volume	0.93 m ³
Pump-down to 54 kPa	4.0 min
Shell collapse factor at full vacuum	6.7 (heads 40, window 7.1 on yield)
Door force at the 5,000 m point	24 kN
Thrust range and error	0 to 50 N, 0.13 N (0.27 % of full scale)
Torque and speed error	0.28 % of 2 N m; 0.02 %
Density error	0.6 %
Chamber effect (estimate)	2.9 % to 5.3 % more thrust than open air with the baffle
Mass	About 470 kg vessel, about 590 kg with pump and cabinet
Cost	Value-engineering target: USD 1,000. Estimated cost of the constructable design: USD 4,920 (USD 3,920 over the target)

AltIRig: energy at one full-power test point, W (ALR-CAL-001 estimates); all of it ends as heat in the steel walls

github.com/BoujeeEnjinia1701/altirig

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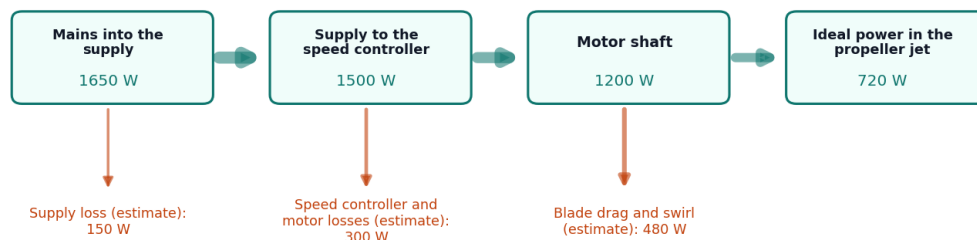


Figure 4. Energy at one full-power test point; every watt ends as heat in the steel walls.

■ Patent design-arounds

From the preliminary patent, trademark and prior-art screen (not legal advice):

- None identified in the preliminary screen.

■ Shared blocks

- Kitewright Lift and Kitewright Range (propulsion under test; they publish their propulsion kits using AltiRig data)
- ColdCell (packs are not tested inside the chamber in the first version; see Safety)
- CalRig proof-load method for calibrating the load cells, if adopted

■ Safety

Safety: AltiRig is a vacuum vessel with a spinning propeller inside, run from mains power.

- **Vacuum (pressure) hazard.** At the 5,000 m point the air outside pushes on the vessel with 39 kPa, about 4 tonnes-force on each square metre. Collapse or a failed window can throw fragments. The vessel is designed for full vacuum with a collapse factor of 6.7, the relief valve stops the chamber below 46.3 kPa, and the first pump-down is a proof test behind a guard with nobody in line with the window. Only polycarbonate is used for the window; acrylic and glass shatter.
- **Door.** The door weighs about 72 kg and is held shut by up to 24 kN. Never force it: vent first through the bleed or vent valve, then unlatch.
- **Spinning propeller.** A propeller can break at speed; fragments fly in its own plane. The window and feed-through are outside that zone, the steel shell contains it, and the motor can only be armed with the door latched and the key turned.
- **Batteries.** No lithium or sodium-ion pack goes inside the chamber in this version. A battery fire in a closed vessel raises the pressure and the fumes are toxic; battery tests need a vent and fire plan that is not part of this design.
- **Mains.** The pump and the motor supply run from mains through a 30 mA residual current device with the vessel earthed. The emergency stop removes motor and pump power, and the bleed valve opens when power is lost.
- **Noise and oil mist.** The pump exhausts oil mist; lead the exhaust outdoors. Wear hearing protection near the running propeller.

AltiRig is published as an open engineering reference. It is not certified pressure equipment or test equipment.

■ Open questions

None are open in the design. Decisions still to be made are kept in the design decisions register (docs/06-design-decisions.md), and items to confirm when parts are bought are listed there.